

Setting up the folder for the lab:

```
joaoribeiro@JOCAPC:~$ mkdir loki-lab && cd loki-lab
joaoribeiro@JOCAPC:~/loki-lab$ docker network rm monitoring
failed to connect to the docker API at unix:///var/run/docker.sock; check if the path is correct and if the daemon is running: dial unix /var/run/docker.sock: connect: no such file or directory
exit status 1
joaoribeiro@JOCAPC:~/loki-lab$ docker login

Authenticating with existing credentials... [Username: joaoribeiro9595]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login did not succeed, error: failed to connect to the docker API at unix:///var/run/docker.sock; check if the path is correct and if the daemon is running: dial unix /var/run/docker.sock: connect: no such file or directory

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: KBWX-PWFF
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
Login Succeeded
joaoribeiro@JOCAPC:~/loki-lab$ sudo service docker start
```

Getting the configuration files for our loki and promtail docker images:

```
joaoribeiro@JOCAPC:~/loki-lab$ wget https://raw.githubusercontent.com/grafana/loki/v3.0.0/cmd/loki/loki-local-config.yaml -O loki-config.yaml
wget https://raw.githubusercontent.com/grafana/loki/v3.0.0/cmd/promtail/promtail-docker-config.yaml -O promtail-config.yaml
--2026-05-21 11:15:15-- https://raw.githubusercontent.com/grafana/loki/v3.0.0/cmd/loki/loki-local-config.yaml
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1285 (1.3K) [text/plain]
Saving to: 'loki-config.yaml'

loki-config.yaml      100%[=====] 1.25K  --.-KB/s  in 0s

2026-05-21 11:15:15 (23.5 MB/s) - 'loki-config.yaml' saved [1285/1285]

--2026-05-21 11:15:15-- https://raw.githubusercontent.com/grafana/loki/v3.0.0/cmd/promtail/promtail-docker-config.yaml
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 298 [text/plain]
Saving to: 'promtail-config.yaml'

promtail-config.yaml 100%[=====] 298  --.-KB/s  in 0s

2026-05-21 11:15:16 (14.6 MB/s) - 'promtail-config.yaml' saved [298/298]
```

Both docker images running and also the docker network:

```
joaoribeiro@JOCAPC:~/loki-lab$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
e2a66a70bc7   grafana/grafana  "/run.sh"               2 minutes ago Up 2 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp  grafana
b42092918aac   busybox        "sh -c 'while true; ..." 7 minutes ago Up 7 minutes  -                                           sample-app
fea2489a5dbd   grafana/promtail:3.0.0  "/usr/bin/promtail -..." 11 minutes ago Up 54 seconds  -                                           promtail
384d695c1c5c   grafana/loki:3.0.0    "/usr/bin/loki -conf..." 12 minutes ago Up 12 minutes  0.0.0.0:3100->3100/tcp, [::]:3100->3100/tcp  loki

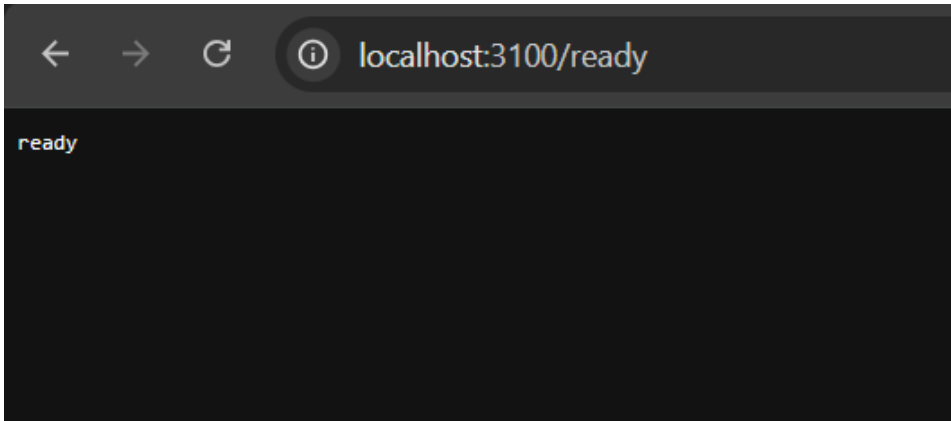
joaoribeiro@JOCAPC:~/loki-lab$ docker network ls | grep loki
docker: unknown command: docker network

Run 'docker --help' for more information
joaoribeiro@JOCAPC:~/loki-lab$ docker network ls | grep loki
ddc7996f8dcd  loki-lab-net  bridge  local
```

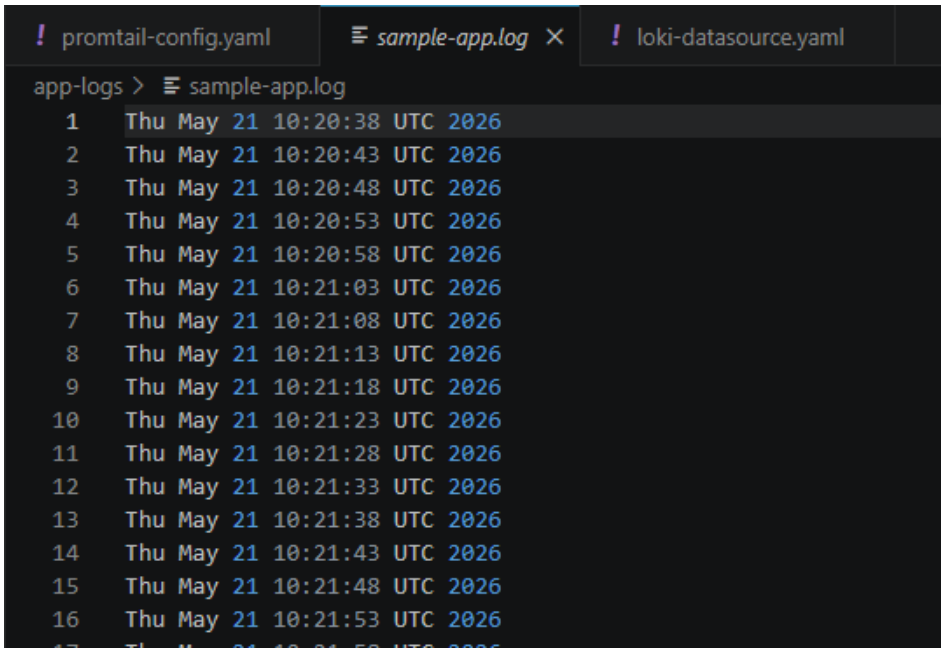
Loki metrics running:

```
← → ↺ ⌕ localhost3100/metrics
# HELP deprecated_flags_inuse_total The number of deprecated flags currently set.
# TYPE deprecated_flags_inuse_total counter
deprecated_flags_inuse_total 0
# HELP go_gc_cpu_to_c_calls_calls_total Count of calls made from Go to C by the current process.
# TYPE go_gc_cpu_to_c_calls_calls_total counter
go_gc_cpu_to_c_calls_calls_total 0
# HELP go_cpu_classes_gc_mark_assist_cpu_seconds_total Estimated total CPU time goroutines spent performing GC tasks to assist the GC and prevent it from falling behind the application. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_gc_mark_assist_cpu_seconds_total counter
go_cpu_classes_gc_mark_assist_cpu_seconds_total 0.40828561
# HELP go_cpu_classes_gc_mark_dedicated_cpu_seconds_total Estimated total CPU time spent performing GC tasks on processors (as defined by GOMAXPROCS) dedicated to those tasks. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_gc_mark_dedicated_cpu_seconds_total counter
go_cpu_classes_gc_mark_dedicated_cpu_seconds_total 0.34879015
# HELP go_cpu_classes_gc_mark_idle_cpu_seconds_total Estimated total CPU time spent performing GC tasks on spare CPU resources that the Go scheduler could not otherwise find a use for. This should be subtracted from the total GC CPU time to obtain a measure of compulsory GC CPU time. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_gc_mark_idle_cpu_seconds_total counter
go_cpu_classes_gc_mark_idle_cpu_seconds_total 0.406823851
# HELP go_cpu_classes_gc_pause_cpu_seconds_total Estimated total CPU time spent with the application paused by the GC. Even if only one thread is running during the pause, this is computed as GOMAXPROCS times the pause latency because nothing else can be executing. This is the exact sum of samples in /go/pauses/seconds; if each sample is multiplied by GOMAXPROCS at the time it is taken. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_gc_pause_cpu_seconds_total counter
go_cpu_classes_gc_pause_cpu_seconds_total 0.0035439
# HELP go_cpu_classes_gc_total_cpu_seconds_total Estimated total CPU time spent performing GC tasks. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics. Sum of all metrics in /cpu/classes/gc.
# TYPE go_cpu_classes_gc_total_cpu_seconds_total counter
go_cpu_classes_gc_total_cpu_seconds_total 0.29735947
# HELP go_cpu_classes_idle_cpu_seconds_total Estimated total available CPU time not spent executing any Go or Go runtime code. In other words, the part of /cpu/classes/totalcpu-seconds that was unused. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_idle_cpu_seconds_total counter
go_cpu_classes_idle_cpu_seconds_total 14217.030412676
# HELP go_cpu_classes_scavenge_assist_cpu_seconds_total Estimated total CPU time spent returning unused memory to the underlying platform in response eagerly in response to memory pressure. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_scavenge_assist_cpu_seconds_total counter
go_cpu_classes_scavenge_assist_cpu_seconds_total 2.71e-07
# HELP go_cpu_classes_scavenge_background_cpu_seconds_total Estimated total CPU time spent performing background tasks to return unused memory to the underlying platform. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_scavenge_background_cpu_seconds_total counter
go_cpu_classes_scavenge_background_cpu_seconds_total 0.003444307
# HELP go_cpu_classes_scavenge_total_cpu_seconds_total Estimated total CPU time spent performing tasks that return unused memory to the underlying platform. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics. Sum of all metrics in /cpu/classes/scavenge.
# TYPE go_cpu_classes_scavenge_total_cpu_seconds_total counter
go_cpu_classes_scavenge_total_cpu_seconds_total 0.003444307
# HELP go_cpu_classes_total_cpu_seconds_total Estimated total available CPU time for user Go code or the Go runtime, as defined by GOMAXPROCS. In other words, GOMAXPROCS integrated over the wall-clock duration this process has been executing for. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics. Sum of all metrics in /cpu/classes.
# TYPE go_cpu_classes_total_cpu_seconds_total counter
go_cpu_classes_total_cpu_seconds_total 14222.40409702
# HELP go_cpu_classes_user_cpu_seconds_total Estimated total CPU time spent running user Go code. This may also include some small amount of time spent in the Go runtime. This metric is an overestimate, and not directly comparable to system CPU time measurements. Compare only with other /cpu/classes/metrics.
# TYPE go_cpu_classes_user_cpu_seconds_total counter
go_cpu_classes_user_cpu_seconds_total 4.71366619
# HELP go_gc_cycles_automatic_gc_cycles_total Count of completed GC cycles generated by the Go runtime.
# TYPE go_gc_cycles_automatic_gc_cycles_total counter
go_gc_cycles_automatic_gc_cycles_total 13
# HELP go_gc_cycles_forced_gc_cycles_total Count of completed GC cycles forced by the application.
# TYPE go_gc_cycles_forced_gc_cycles_total counter
go_gc_cycles_forced_gc_cycles_total 0
# HELP go_gc_cycles_total_gc_cycles_total Count of all completed GC cycles.
# TYPE go_gc_cycles_total_gc_cycles_total counter
go_gc_cycles_total_gc_cycles_total 13
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 4.8721e-05
```

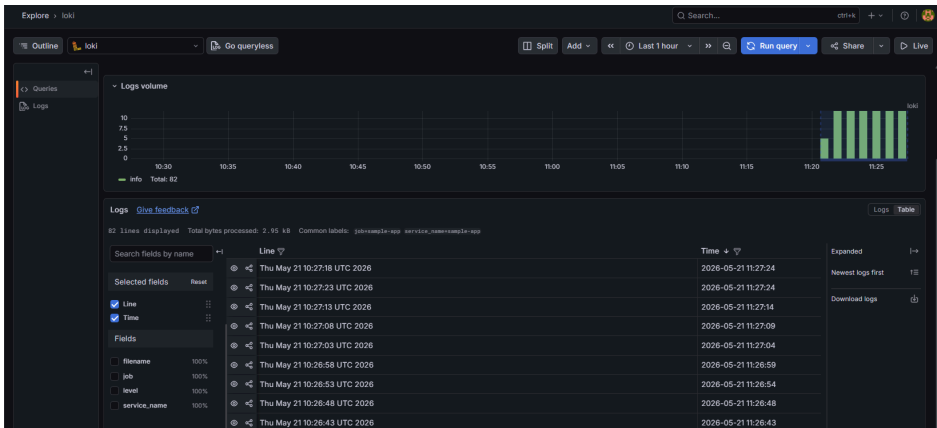
Loki is ready for action:



App logs running:



Also in the Grafana Explorer:



Loki Dashboard from grafana:

